Enveda CASMI 2026 - Molecule ID From Mass Spectra

Predict 2D chemical structures of small molecules detected in complex biological extracts

TASK	EVALUATION METRIC	SUBMISSION MODE	COMPUTE LIMIT
MS/MS → SMILES	MRR@25 (Connectivity)	Kaggle Notebooks Only	≤ 9 hrs CPU / GPU

1. Description

Modern mass spectrometry can detect thousands of molecules in nature, but identifying them all remains a challenge. Researchers must predict chemical structures from tandem mass spectra, even for molecules never seen before.

Current methods mostly compare spectra against reference libraries. They work for known compounds but struggle with the many unknown molecules found in real biological samples. Many still require slow, expensive lab work to identify.

In this competition, you'll build machine learning models that predict 2D chemical structures (a SMILES string) from LC-MS/MS spectra. Your goal is to generate accurate SMILES representations for both known and novel molecules.

Your solution could help researchers discover new medicines and identify disease biomarkers.

2. Background — Principles of Mass Spectrometry

What is an MS/MS mass spectrum?

You don't need a chemistry background to compete, but it helps to understand the basics of how mass spectra are generated in a mass spectrometer:

- **Ionization:** The instrument can only detect and measure charged ions, so first a molecule is ionized, picking up or losing charged species. In the instrument's **positive ion mode** (more common for small molecules), the resulting ion is positively charged, in **negative ion mode** (less common but still used), it is negatively charged. The kind of charged ion the molecule acquires is called an adduct. For some common examples, the molecule may acquire a proton (positive), acquire an ammonium ion (positive), or lose a proton (negative). We would write these **adduct** forms respectively as `[M+H]+`, `[M+NH4]+`, or `[M−H]−`.
- **Precursor mass detection:** The instrument measures the ion's **precursor m/z** (mass-to-charge ratio) with high accuracy, which strongly constrains the molecular formula. For small molecules, the charge (z) is usually +1 or -1, so m/z corresponds to the mass of the ion, and sometimes we will refer to m/z simply as "mass".
- **Fragmentation:** The precursor ion (actually many individual ions of the same molecule+adduct) is then selected and fragmented by collision with neutral gas at a given **collision energy**, and the instrument records the **m/z and intensity (abundance) of each fragment**. The histogram of intensities across all detected fragments is the molecule's mass spectrum. Intensities are typically normalized. A mass spectrum looks like this (see Figure 1 on next page).

2. Background (Continued) — Spectrum & Terminology

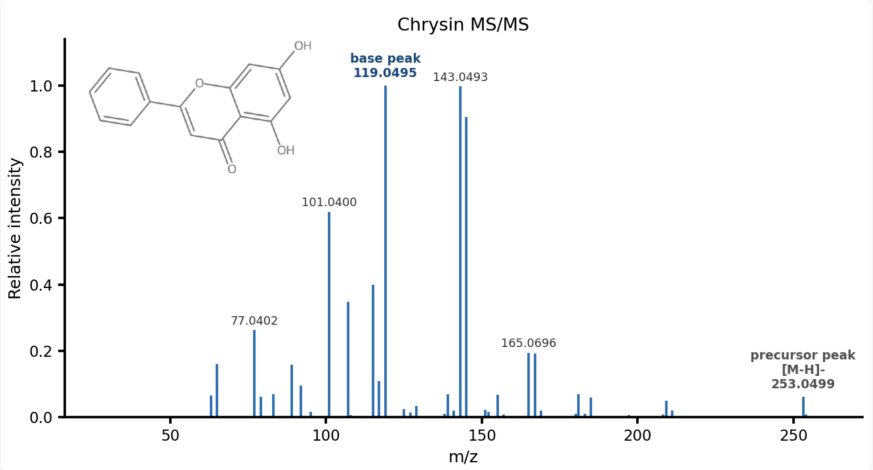


Figure 1: MS/MS spectrum of Chrysin [M-H]- adduct at around 50eV, timsTOF (Source: Enveda)

Fragment Ion / Peak

Each individual *m/z* detected after precursor collision and dissociation.

Base Peak

The highest-intensity fragment ion in the spectrum, representing the most stable or favored fragment.

Precursor Peak

The *m/z* corresponding to the intact, unfragmented ion. It may or may not be present, depending on how thoroughly the ion was fragmented.

3. Evaluation — Metric (MRR@25)

Submissions are evaluated using **Mean Reciprocal Rank @ 25 (MRR@25)**:

$$\text{MRR@25} = \frac{1}{U}\sum_{u=1}^{U}\frac{1}{\text{rank}_u}$$

where U is the number of molecules and $rank_u$ is the position of the first correct structure in your ranked list for molecule u.

A molecule scores 0 if none of your guesses is correct. Each molecule has exactly one correct structure, so only your **first** correct guess counts. A correct guess at position 1 scores 1.0, at position 2 scores 0.5, at position 25 scores 0.04.

Candidate Rank	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	...	Rank 25	Not in Top 25
Reciprocal Score	**1.000**	**0.500**	**0.333**	**0.250**	**0.200**	...	**0.040**	**0.000**

3. Evaluation (Continued) — Matching & Submission File

Matching: Atom Connectivity & InChIKey14

A prediction is correct when it describes the same **atom connectivity** as the answer. Both your SMILES and the answer are passed through RDKit's tautomer canonicalization (pinned at `2026.03.3`) and reduced to the first block of their InChIKey (the **InChIKey14**), then compared. The prediction is correct when the two first-block keys match.

> **Stereocenters and Tautomers:** You are *not* penalized for getting stereocenters or tautomer forms wrong. For example, both of the following score identically against an answer of glucose, because they reduce to the same InChIKey14 (`WQZGKKKJIJFFOK`):
>
> ```
> OC[C@H]1OC(O)[C@H](O)[C@@H](O)[C@@H]1O
> OCC1OC(O)C(O)C(O)C1O
> ```

Official Metric Notebook: CASMI Mean Reciprocal Rank Notebook.

Submission File Format

For each `molecule_id` in the test set, predict up to 25 candidate structures as SMILES, **best guess first**, joined by semicolons in a single field. Every `molecule_id` must appear exactly once. The file should contain a header and have the following format:

```
molecule_id,smiles
m_0014ef,CC1=CC(=O)C=CC1=O;OC(=O)c1ccccc1O;CN1C=NC2=C1C(=O)N(C)C(=O)N2C
m_004d06,NCCc1ccc(O)cc1;CC(=O)Nc1ccc(O)cc1;OCC(O)CO
...
```

Fewer than 25 guesses is allowed but there is no penalty for a wrong guess beyond the rank it occupies.

> **Submission Validation Rules:** A submission is rejected if it is missing the `molecule_id` or `smiles` column, is empty, contains nulls in either column, repeats a `molecule_id`, or gives more than 25 semicolon-separated guesses for any molecule.

4. Code Requirements



Submissions to this competition must be made through **Kaggle Notebooks**. In order for the "Submit" button to be active after a commit, the following conditions must be met:

- **CPU Notebook:** ≤ 9 hours run-time
- **GPU Notebook:** ≤ 9 hours run-time
- **Internet access:** Disabled during scoring
- **External data:** Freely & publicly available external data is allowed, including pre-trained models
- **Submission file:** Must be named `submission.csv`

For guidance, see the Kaggle Code Competition FAQ and the Code Debugging Guide.

5. Acknowledgements

CASMI: The Critical Assessment of Small Molecule Identification (CASMI) was founded in 2012 by Emma Schymanski and Steffen Neumann, and run between 2012 and 2022 by various academic teams. We revive the name with their permission and are grateful for their support.